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Centre Number

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Student Number

SCEGGS Darlinghurst

2007

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Diagrams should be drawn in pencil
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start each question in a new booklet

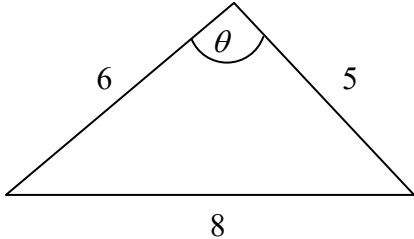
Total marks – 120

- Attempt Questions 1–10
- All questions are of equal value

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Total marks – 120
Attempt Questions 1–10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

	Marks
Question 1 (12 marks)	
(a) Find the exact value of $\sqrt{1\frac{32}{49}}$.	1
(b) Solve $(x - 5)^2 = 16$.	2
(c) (i) State the domain of the function $y = \frac{3}{2 - x}$.	1
(ii) Sketch the graph of $y = \frac{3}{2 - x}$ showing all important features.	2
(d) Find the value of θ in the diagram. Give your answer correct to the nearest minute.	2
 NOT TO SCALE	
(e) Express $3\sqrt{5} + \sqrt{20}$ in the form $\sqrt{a}$.	2
(f) Over 7 years, \$125 grows to \$165. Interest is compounded annually. Find the compound interest rate as a percentage per annum. Give your answer correct to one decimal place.	2

Question 2 (12 marks) Use a SEPARATE writing booklet.

(a) Differentiate:

(i) $\cos(3x^2 + 1)$ **2**

(ii) $\frac{x - e^{2x}}{e^x}$ **2**

(b) Find a primitive of $\sqrt[3]{(2x+1)^2}$. **2**

(c) Evaluate $\int_2^3 \frac{x}{x^2 - 1} dx$ **3**

Give your answer correct to 3 significant figures.

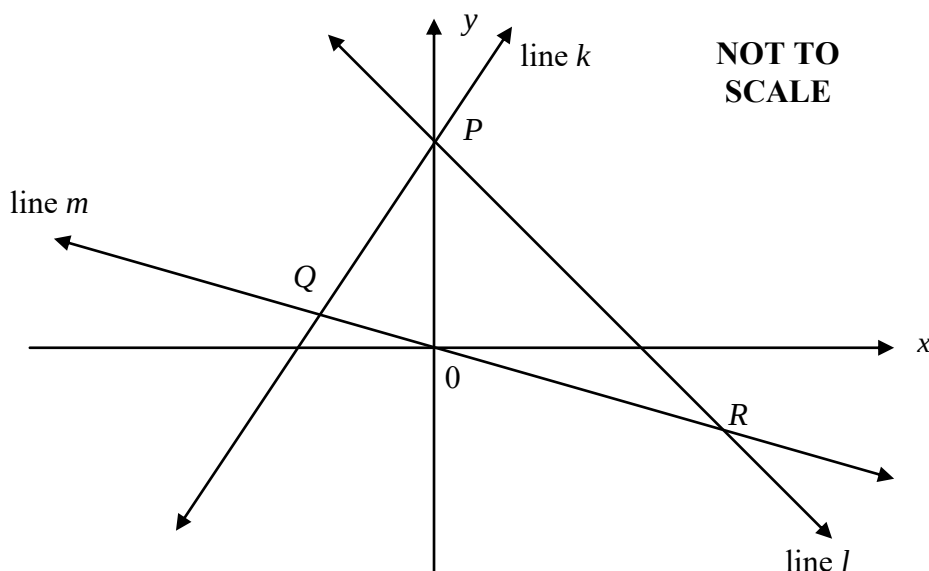
(d) For a function $g(x)$ it is given that **3**

$g'(x) = 3x^2 - 4 + \frac{1}{x^2}$ and $g(x) = 4$ when $x = 1$.

Find the equation $g(x)$.

Question 3 (12 marks) Use a SEPARATE writing booklet.

(a)



The point $Q(-2, 1)$ lies on the line k whose equation is $9x - 2y + 20 = 0$.
The point $R(4, -2)$ lies on the line l whose equation is $3x + y - 10 = 0$.

- (i) By solving simultaneously, find the point P where k and l intersect. 2
- (ii) Find the equation of the line m which joins Q and R . 2
- (iii) Show that the exact perpendicular distance from P to line m is $4\sqrt{5}$ units. 2
- (iv) Hence, or otherwise, find the exact value of the area of the triangle bounded by the three lines k , l and m . 2

(b) For the arithmetic progression $81, 77, 73, \dots$

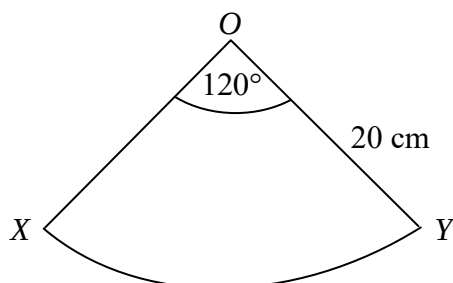
- (i) show that the sum to n terms is given by the expression 1

$$S_n = 83n - 2n^2$$

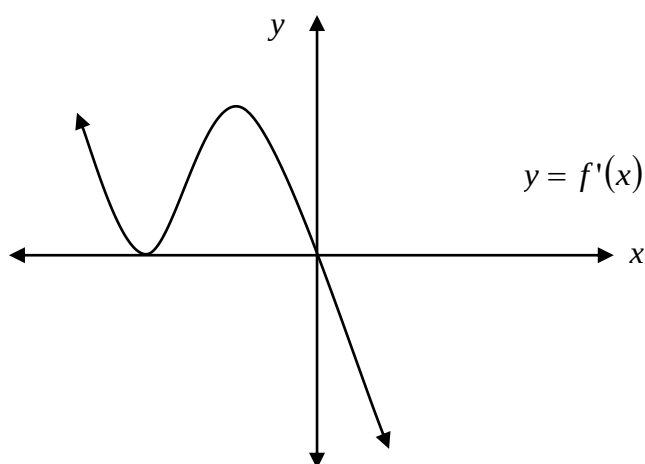
- (ii) Find the smallest value of n for which the sum to n terms would be negative. 3

Question 4 (12 marks) Use a SEPARATE writing booklet.

- (a) Francesca wants to design a conical party hat for her 18th birthday party. She folds the sector XOY so that the edges OX and OY coincide to form a cone.



- (i) Find the exact length of the arc XY . 1
- (ii) Find the radius of the base of the cone formed. 1
- (b) Find the equation of the normal to the curve $y = \sin\left(2x + \frac{\pi}{2}\right)$ at the point 3
where $x = \frac{\pi}{4}$.
- (c) The diagram below shows the first derivative $f'(x)$ of a function. 3
Copy the diagram into your answer booklet and on the same axes draw a possible curve for the function $f(x)$.



Question 4 continues on page 6

Question 4 (continued)

- (d) A box contains 10 chocolates all of identical appearance.
Four have caramel centres and the other six have mint centres.
Jolene randomly selects and eats three chocolates from a box.

Find the probability that Jolene eats:

- | | |
|-------------------------------------|----------|
| (i) three mint chocolates. | 1 |
| (ii) exactly one caramel chocolate. | 2 |
| (iii) at least one mint chocolate. | 1 |

Question 5 (12 marks) Use a SEPARATE writing booklet.

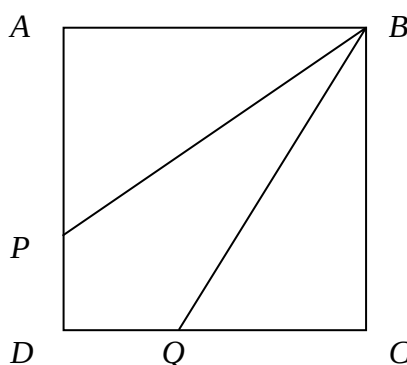
- (a) A quadratic equation with roots α and β has the form

2

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

Hence, or otherwise, form a quadratic equation whose roots are $3 + \sqrt{5}$ and $3 - \sqrt{5}$.

- (b)



**NOT TO
SCALE**

$ABCD$ is a square. P and Q are points on AD and DC respectively such that $PD = QD$.

Copy the diagram into your answer booklet.

- (i) Prove that $\triangle BAP$ is congruent to $\triangle BCQ$.

3

- (ii) If $\frac{BP}{BA} = \frac{3}{2}$ find the exact value of $\tan \angle APB$.

2

- (c) Luigi decides to set up a trust fund for his grand-daughter Sophia. He plans to give it to her on her 21st birthday. He invests \$150 at the beginning of each month. The money is invested at 9% per annum, compounded monthly.

The trust fund matures at the end of the month of his final investment, 21 years after his first investment. This means that Luigi makes 252 monthly investments.

- (i) After 21 years, what will be the value of the first \$150 invested?
- (ii) By writing a geometric series for the value of all Luigi's investments, the final value of Sophia's trust fund.

2

3

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the function defined by $f(x) = x^3(2 - x)$.
- (i) Find the coordinates of the stationary points and determine their nature. **3**
 - (ii) Find the coordinates of any point of inflexion. **2**
 - (iii) Sketch the graph of $y = f(x)$ for $-1 \leq x \leq 2$. **3**
 - (iv) For the given domain $-1 \leq x \leq 2$ when is the curve concave up? **1**
- (b) Calculate the exact volume generated when the arc of the curve **3**

$$y = 2 \sec \frac{x}{2}$$

between $x = \frac{\pi}{3}$ and $x = \frac{\pi}{2}$ is rotated about the x axis.

Question 7 (12 marks) Use a SEPARATE writing booklet.

- (a) Solve the equation: 2

$$9^x - 4 \cdot 3^x + 3 = 0$$

- (b) The point $P(x, y)$ moves such that its distance from the point $R(-1, 0)$ is always twice its distance from the point $S(2, 0)$.

- (i) Show that the equation of the locus of point P is given by 2
 $x^2 - 6x + y^2 + 5 = 0$.

- (ii) Describe the locus geometrically. 2

- (c) For the curve $y = 2 \sin x - 1$.

- (i) Find the roots of the equation $2 \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$. 2

- (ii) Sketch the curve for $0 \leq x \leq 2\pi$ showing all important features and points of intersection with the x axis. 2

- (iii) Find the area under the curve that lies above the x axis in the given domain. 2

Question 8 (12 marks) Use a SEPARATE writing booklet.

- (a) (i) For what values of x will the geometric progression 1

$$1 + (x - 2) + (x - 2)^2 + \dots$$

have a limiting sum?

- (ii) If this series has a limiting sum of 2, find the value of x . 2

- (b) Consider the curve $f(x) = \frac{e^x}{x}$.

- (i) State the domain of $y = f(x)$. 1

- (ii) Show that $f'(x) = \frac{e^x(x-1)}{x^2}$. 1

- (iii) Find the co-ordinates of the stationary point and determine its nature. 3

- (iv) Explain why $y = f(x)$ has no x -intercepts. 1

- (v) What happens to $f(x)$ as $x \rightarrow -\infty$. 1

- (vi) Sketch the curve $y = f(x)$, showing all important features. 2

Question 9 (12 marks) Use a SEPARATE writing booklet.

- (a) Find the co-ordinates of the vertex and the focus of the parabola 2

$$y = x^2 - 6x + 10$$

- (b) (i) Use Simpson's rule with five function values to evaluate $\int_1^3 5^{2x} dx$ 2
correct to three decimal places.

- (ii) The region bounded by the curve $y = 5^x$, the lines $x = 1$ and $x = 3$ 2
is rotated about the x -axis. Using part (i) or otherwise, find an
estimate for the volume of revolution formed.
Answer correct to three decimal places.

- (c) Luke has made up a new game for one person that is played with two dice.
He rolls both dice and if he rolls a difference of 0 or 1 he wins but if he rolls
a difference of 4 or 5 he loses. Any other difference means he rolls the dice again.

- (i) What is the probability that Luke will win on his first roll of the dice? 1

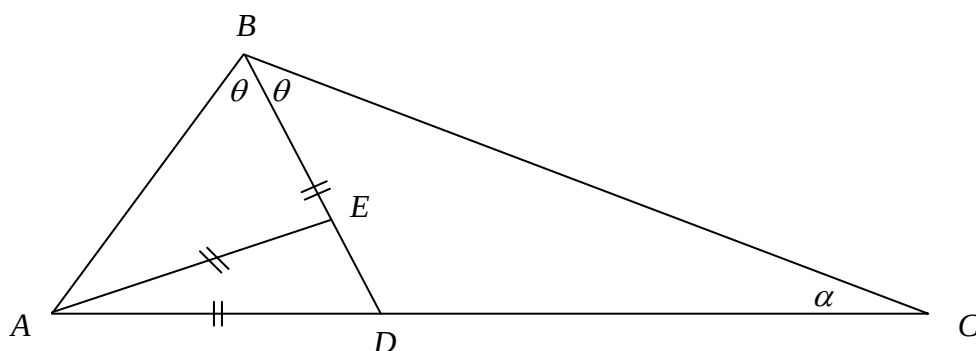
- (ii) Calculate the probability that a second throw is needed. 1

- (iii) What is the probability that Luke wins on his first, second or third
throw? Leave your answer unsimplified. 2

- (iv) Calculate the probability that Luke wins the game. 2

Question 10 (12 marks) Use a SEPARATE writing booklet.

(a)



In the diagram above, ABD and AED are isosceles triangles with $AD = BD = AE$, and BD bisects $\angle ABC$. Let $\angle ABC = \angle CBD = \theta$ and let $\angle DCB = \alpha$.

- (i) Show that $\angle EAB = \alpha$, giving reasons. 3
- (ii) Hence show that $\triangle ABE \parallel \triangle CBD$. 1
- (iii) Deduce that $AE^2 = BE \times CD$. 2

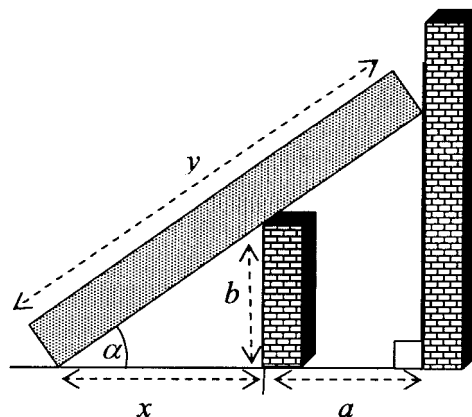
Question 10 continues on page 13

Question 10 (continued)

- (b) Devendra is a motorcycle stunt bike rider in an upcoming Bollywood feature film.

He will be required to ride from a wall onto a beam, which passes over a second lower wall b metres high and located a metres from the first wall.

Let the length of the beam by y metres, the angle the beam makes with the horizontal be α and x the distance from the foot of the beam to the smaller wall.



- (i) Show that $y = a \sec \alpha + b \operatorname{cosec} \alpha$. 2

- (ii) Show that 1

$$\frac{dy}{d\alpha} = \frac{-b \cos^3 \alpha + a \sin^3 \alpha}{\sin^2 \alpha \cos^2 \alpha}$$

- (iii) Hence show that if $\frac{dy}{d\alpha} = 0$ then $\tan \alpha = \sqrt[3]{\frac{b}{a}}$. 1

- (iv) Hence show that the shortest beam that can be used is given by: 2

$$y = a \sqrt{1 + \left(\frac{b}{a}\right)^{\frac{2}{3}}} + b \sqrt{1 + \left(\frac{a}{b}\right)^{\frac{2}{3}}}$$

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

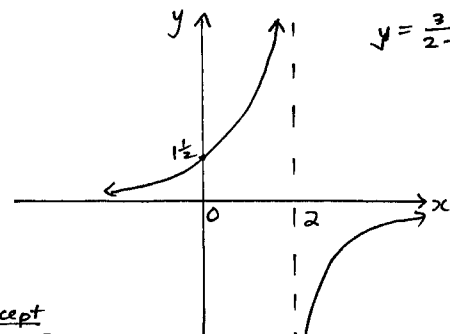
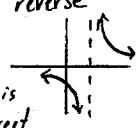
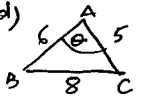
$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Q1 a) $1\frac{2}{7}$ ✓	Q1: Comm/2
b) $(x-5)^2 = 16$ $x-5 = \pm 4$ $x = 5 \pm 4$ $x = 5-4 = 1$ $x = 5+4 = 9$ ✓✓	remember $\sqrt{16} = \pm 4$ so there will be two solutions
c) i) $y = \frac{3}{2-x}$ <u>domain</u> all real x except $x = 2$. ✓	you must include "all real x"....
ii)  $y = \frac{3}{2-x}$ y-intercept when $x=0$ $y = \frac{3}{2}$	Must show • asymptote • y-intercept • correct shape • Not well done. Most had the sketch in reverse • if you find the y-int it is obvious that this cannot be correct.  Comm 2
d)  $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ $\cos \theta = \frac{5^2 + 6^2 - 8^2}{2 \times 5 \times 6}$ $\cos \theta = -0.05$ $\theta = 92^\circ 52'$ (to nearest minute.) ✓	Learn your formula
e) $3\sqrt{5} + \sqrt{20}$ $= 3\sqrt{5} + 2\sqrt{5}$ $= 5\sqrt{5}$ $= \frac{\sqrt{25} \times 5}{1}$ $= \sqrt{125}$ ✓✓	most were fine with $5\sqrt{5}$ but many did not write in the form $\sqrt{a}$
f) $A = 165$ $P = 125$ $n = 7$ $r = ?$ using compound interest formula. $A = P(1 + \frac{r}{100})^n$ $165 = 125(1 + \frac{r}{100})^7$ solve for r $1.32 = (1 + \frac{r}{100})^7$ $1 + \frac{r}{100} = \sqrt[7]{1.32}$ $\frac{r}{100} = \sqrt[7]{1.32} - 1$ $r = (\sqrt[7]{1.32} - 1) \times 100$ $r \approx 4.0\%$ p.a. (1 d.p.) ✓	Use the compound interest formula! Not series.

Q2 a) i) $\frac{d}{dx}(\cos(3x^2+1))$ $= -\sin(3x^2+1) \times 6x$ $= -6x \sin(3x^2+1)$ ✓✓	Q2: Calc /12 $\frac{d}{dx}(\cos f(x)) = -\sin f(x) \times f'(x)$ Do not attempt to use the product rule as it is only in one place. Calc/2
ii) $y = \frac{x-e^{2x}}{e^x}$ $u = x - e^{2x}$ $v = e^x$ $u' = 1 - 2e^{2x}$ $v' = e^x$ Using the Quotient rule $y' = \frac{vu' - uv'}{v^2}$ $= \frac{e^x(1-2e^{2x}) - e^x(x-e^{2x})}{(e^x)^2}$ ✓ $= \frac{e^x - 2e^{3x} - xe^x + e^{3x}}{e^{2x}}$ $= \frac{e^x - e^{3x} - xe^x}{e^{2x}}$ $= \frac{e^x(1 - e^{2x} - x)}{e^{2x}}$ $= \frac{1 - e^{2x} - x}{e^x}$	It is better to set your work out like this rather than trying to do the steps in your head. You only need to show you know how to use the quotient rule to get this mark. You must include the brackets. Calc/2 Simplification not necessary in this question.
b) $\int \sqrt[3]{(2x+1)^2} dx$ $= \int (2x+1)^{2/3} dx$ $= \frac{(2x+1)^{5/3}}{\frac{5}{3} \times 2} + C$ $= \frac{3}{10} (2x+1)^{5/3} + C$ ✓ must have +C ✓	This part was not very well done. It is an easy question. Please learn how to use the chain rule and how to change into index form. $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$ Calc/2

②

$$\int_2^3 \frac{x}{x^2-1} dx$$

$$= \frac{1}{2} \int_2^3 \frac{2x}{x^2-1} dx$$

$$= \frac{1}{2} [\log_e(x^2-1)]_2^3$$

$$= \frac{1}{2} (\log_e 8 - \log_e 3)$$

$$= \frac{1}{2} \log_e \left(\frac{8}{3}\right)$$

$$\approx 0.490 \quad (3 \text{ s.f.})$$

$$\int \frac{f'(x)}{f(x)} dx = \log_e f(x) + C$$

You should recognise that this is a logarithm.

$$\text{Use the log. law} \\ \log_a - \log_b = \log \frac{a}{b}$$

or just use your calculator using the $\ln$ key.

Calc 3

d) $g'(x) = 3x^2 - 4 + \frac{1}{x^2}$
 $= 3x^2 - 4 + x^{-2}$

$$g(x) = \int (3x^2 - 4 + x^{-2}) dx$$

$$= \frac{3x^3}{3} - 4x + \frac{x^{-1}}{-1} + C$$

$$= x^3 - 4x - \frac{1}{x} + C$$

(when $x=1$, $g(x)=4$)

$$4 = 1 - 4 - \frac{1}{1} + C$$

$$4 = -4 + C$$

$$\therefore C = 8$$

Equation of $g(x)$

$$g(x) = x^3 - 4x - \frac{1}{x} + 8$$

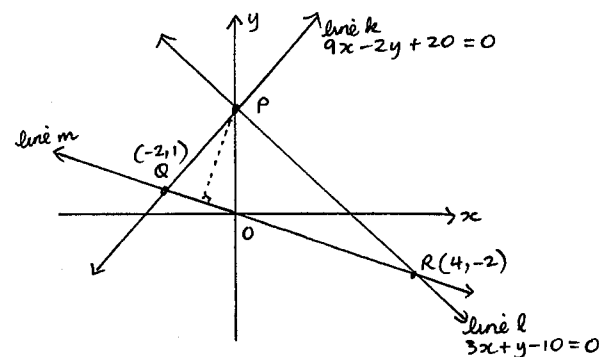
You will have more success if you write all terms in index form before attempting to integrate.

Don't forget to include $+C$ in this question.

This is a standard question. Make sure you practise this type of integration.

Calc 3

Q3
a



Q3: Reas / 3

$$\begin{array}{l} \text{i) } 9x - 2y + 20 = 0 \quad (1) \\ \quad 3x + y - 10 = 0 \quad (2) \end{array}$$

$$\begin{array}{l} (2) \times 2 \quad 6x + 2y - 20 = 0 \quad (3) \\ \quad 9x - 2y + 20 = 0 \quad (1) \end{array}$$

Add (3) & (1)

$$\begin{array}{l} 15x = 0 \\ x = 0 \end{array}$$

Subst. into (1)

$$\begin{array}{l} -2y + 20 = 0 \\ 2y = 20 \\ y = 10 \end{array}$$

$\therefore P$ has coordinates $(0, 10)$

ii) $Q(-2, 1) \quad R(4, -2)$

$$m_{QR} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-2 - 1}{4 - (-2)}$$

$$= \frac{-3}{6}$$

$$= -\frac{1}{2}$$

Equation of QR

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -\frac{1}{2}(x + 2)$$

$$2y - 2 = -x - 2$$

$$x + 2y = 0$$

Well Done.

Well Done

Q3 a) cont'.
 iii) $P(0,10)$ line m $x+2y=0$
 $pd = \frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$
 $= \frac{|0+2 \times 10|}{\sqrt{1^2+2^2}}$
 $= \frac{20}{\sqrt{5}}$
 $= \frac{20}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$
 $= \frac{20\sqrt{5}}{5}$
 $= 4\sqrt{5}$ units.

Well Done.

- you must show the process of rationalising the denominator not just go from $\frac{20}{\sqrt{5}}$ to $4\sqrt{5}$
- this is necessary as $4\sqrt{5}$ is given to you

iv) $QR = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$
 $= \sqrt{(-2-4)^2 + (1+2)^2}$
 $= \sqrt{(-6)^2 + 3^2}$
 $= \sqrt{36+9}$
 $= \sqrt{45}$
 $= 3\sqrt{5}$

Well Done

$\therefore \text{Area } \Delta = \frac{1}{2} \times \text{base} \times \text{perp. height}$
 $= \frac{1}{2} \times 3\sqrt{5} \times 4\sqrt{5}$
 $= \frac{1}{2} \times 12 \times 5$
 $= 30$ units²

Careless errors with multiplication

Q3 b) AP 81, 77, 73, ...

$a=81$
 $d=-4$

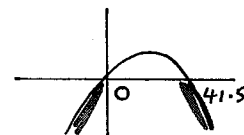
i) $S_n = \frac{n}{2} [2a + (n-1)d]$
 $= \frac{n}{2} [162 + (n-1) \times -4]$
 $= \frac{n}{2} [162 - 4n + 4]$
 $= \frac{n}{2} [166 - 4n]$
 $= 83n - 2n^2$

Well Done

Q3 b) ii) $S_n < 0$

$83n - 2n^2 < 0$

$n(83-2n) < 0$



$n < 0, n > 41.5$

$n < 0$ is not a valid solution

$\therefore n > 41.5$

$n = 42$

• This is a quadratic inequality, you MUST do a sketch to solve this

• MUST state that $n < 0$ is not a valid solution

• Answer the question asked: $n = 42$

Reas 13

Q4 a) $\theta = 120^\circ$

$= \frac{2\pi}{3}$ radians

$l = r\theta$

$XY = 20 \times \frac{2\pi}{3}$

$= \frac{40\pi}{3}$ cm



the points XY join up to form the base of the hat, so the arc XY is the circumference of the base of the hat.

$2\pi r = \frac{40\pi}{3}$

$r = \frac{40\pi}{3} \div 2\pi$

$r = \frac{20}{3}$ cm

Q4: Reas 11
 Comm 13
 Case 13

- done very well

- leave expression as a improper fraction

- this was done poorly and this was due to mainly candidates not knowing the formula for the circumference of a circle.

- there was also many silly errors

Reas 1

(Q4) b) $y = \sin(2x + \frac{\pi}{2})$
 $y' = \cos(2x + \frac{\pi}{2}) \times 2$
 $= 2 \cos(2x + \frac{\pi}{2})$

when $x = \frac{\pi}{4}$

tangent $m_1 = 2 \cos(2x\frac{\pi}{4} + \frac{\pi}{2})$
 $= 2 \cos(\frac{\pi}{2} + \frac{\pi}{2})$
 $= 2 \cos \pi$
 $= -2$

normal $m_2 = \frac{1}{2}$ ✓

when $x = \frac{\pi}{4}$

$y = \sin(2x\frac{\pi}{4} + \frac{\pi}{2})$
 $= \sin \pi$
 $= 0$ $(\frac{\pi}{4}, 0)$ ✓

Equation of normal

$y - y_1 = m(x - x_1)$

$y - 0 = \frac{1}{2}(x - \frac{\pi}{4})$

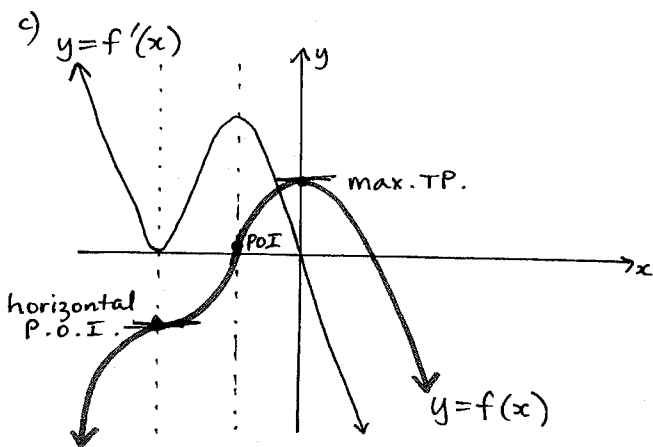
$y = \frac{1}{2}x - \frac{\pi}{8}$ ✓

$x - 2y - \frac{\pi}{4} = 0$

- done fairly well by most candidates but there are still some who need to practice working with radians

- it was pleasing to see nearly all students know the relationship between gradients and perpendicular lines.

(Calc/3)



Slope diagram.

+ 0 + 0 -

/ - / - \

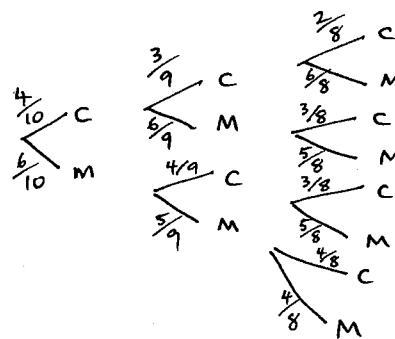
Horizontal POI ✓
 Max TP. ✓
 concavity correct ✓

- most common mistake was sketching the derivative of the graph!

- It is really important students try to be as neat as possible with their graphs i.e. line up where the max tp's and p.o.i. AND LABEL!!!

(Comm/3)

(4) d)



i) $P(MMM) = \frac{6}{10} \times \frac{5}{9} \times \frac{4}{8}$
 $= \frac{1}{6}$ ✓

ii) $P(\text{exactly one C})$

$= P(CMM) + P(MCM) + P(MMC)$

$= (\frac{4}{10} \times \frac{6}{9} \times \frac{5}{8}) \times 3$ ✓ ✓

$= \frac{1}{2}$

iii) $P(\text{at least one M})$

$= 1 - P(\text{no M})$

$= 1 - P(CCC)$

$= 1 - (\frac{4}{10} \times \frac{3}{9} \times \frac{2}{8})$

$= \frac{29}{30}$ ✓

- Done fairly well by candidates although quite a few didn't realise there were no repeats

- Quite a few students didn't realise there was 3 outcomes here NOT 2 outcomes.

- Done fairly well.

(9)

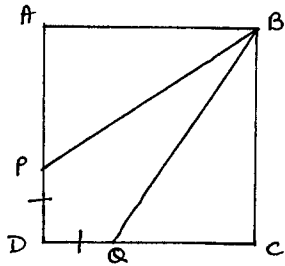
Q5: Reas/7
Comm/3

Very poorly done, despite the fact that all the required theory was given in the question!

Q5 a) Sum of roots.
 $\alpha + \beta = 3 + \sqrt{5} + 3 - \sqrt{5} = 6$
Product of roots
 $\alpha\beta = (3 + \sqrt{5})(3 - \sqrt{5}) = 9 - 5 = 4$

$\therefore$ Equation $x^2 - 6x + 4 = 0$ Must have = 0 ✓ ✓

b)



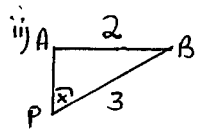
Given
 ABCD is a square
 PD = QC

i) In $\triangle BAP$ and $\triangle BCQ$

$AB = BC$ (equal sides of a square)
 $\angle BAP = \angle BCQ = 90^\circ$ (ABCD is a square) ✓

Since $AD = DC$ (sides in a square)
 and $PD = QC$ (given)
 $\therefore AP = AD - PD$
 $QC = DC - QC$
 $\therefore AP = QC$

$\therefore \triangle BAP \equiv \triangle BCQ$ (SAS) ✓



$\frac{BP}{BA} = \frac{3}{2}$

Find AP by Pythagoras.

$AP^2 = 3^2 - 2^2$

$AP = \sqrt{5}$

$\therefore \tan \angle APB = \frac{2}{\sqrt{5}}$ ✓

This particular reason was not well explained.

Many forgot this, or gave the incorrect test.

(Comm/3)

No one picked up on the fact that AP turned out to be longer than the side of the square!

... other than that it was well done.

(Reas/2)

(10)

Q5 c) $P = \$150$

$r = 9\% \text{ pa}$
 $= \frac{9}{12} \% \text{ per month}$
 $= 0.75\% \text{ per month}$

$n = 21 \text{ years}$
 $= 252 \text{ months}$

Let A be the amount the investment becomes.

The first investment becomes

i) $A_1 = 150 (1.0075)^{252}$
 $= \$985.93$

ii) The second investment becomes

$A_2 = 150 (1.0075)^{251}$

The third investment becomes

$A_3 = 150 (1.0075)^{250}$

$\vdots$

The last investment is

$A_{252} = 150 (1.0075)$

The total investment

$A = 150 [1.0075^{252} + 1.0075^{251} + \dots + 1.0075]$ ✓

GP $a = 1.0075$
 $r = 1.0075$
 $n = 252$
 $S_n = \frac{a(r^n - 1)}{r - 1}$

$= 150 \times \frac{1.0075 \times (1.0075^{252} - 1)}{(1.0075 - 1)}$ ✓

$= \$112,292.96$ ✓

✓ This was a standard "Superannuation" - type question that was very poorly done!

✓ Note that questions about money should be answered to the nearest cent

✓ Note that if you decide to make $a = 1.0075^{252}$ then $r = \frac{1}{1.0075}$

(Reas/5)

(11)

6 a) i) $f(x) = x^3(2-x)$

$$f(x) = 2x^3 - x^4$$

i) $f'(x) = 6x^2 - 4x^3$

$$f''(x) = 12x - 12x^2$$

Stationary points $f''(x) = 0$

$$6x^2 - 4x^3 = 0$$

$$2x^2(3 - 2x) = 0$$

$$x = 0 \quad 3 - 2x = 0$$

$$2x = 3$$

$$x = \frac{3}{2}$$

when $x = 0$
 $y = 0$ (0,0)

when $x = \frac{3}{2}$
 $y = 2x(\frac{3}{2})^3 - (\frac{3}{2})^4$
 $= 1\frac{11}{16}$
 $= 1.6875$ $(1\frac{1}{2}, 1\frac{11}{16})$

Test nature

when $x = 0$
 $y'' = 0$

$\therefore$ possible horizontal point of inflexion.

Check concavity change $y'' = 12x - 12x^2$

$x = -1$	$x = 0$	$x = \frac{1}{2}$
-24	0	3
-	0	+

$\therefore$ Concavity changes, so there is a horizontal point of inflexion at (0,0)

when $x = \frac{1}{2}$

$$y'' = 12x\frac{1}{2} - 12x(\frac{1}{2})^2$$

$$= -9$$

$$y'' < 0 \text{ concave down}$$

$\therefore$ A maximum turning point at $(1\frac{1}{2}, 1\frac{11}{16})$

Q6: Comm/3
Calc/8

Overall, quite poorly done.

Incorrect application of the product rule to differentiate $x^3(2-x)$ caused many grief! Note that it is MUCH easier to expand first

Many left part (i) here with the answer "(0,0)" is a possible horizontal POI. While this is true it doesn't answer the question. You must check, here in part i, if it is or is not a POI.

Also, when determining the nature of an S.P or checking concavity changes you must substitute actual values, simply writing + or - is not enough.

Calc/3

(12)

6 a) ii) Points of inflexion $f''(x) = 0$

$$12x - 12x^2 = 0$$

$$12x(1-x) = 0$$

$$x = 0 \quad x = 1$$

Horizontal point of inflexion at (0,0) from part (i)

when $x = 1$
 $y = 2 - 1$
 $= 1$ (1,1)

check concavity change $y'' = 12x - 12x^2$

$x = \frac{1}{2}$	$x = 1$	$x = 1\frac{1}{2}$
3	0	-9
+	0	-

must have values

Since concavity changes, there is a point of inflexion at (1,1)

- Many forgot to check for a change in concavity

- Note that the number you choose to substitute here

$$x = ? \quad x = 1$$

has to be between 0 & 1

Calc/2

iii) Endpoints

$$x = -1$$

$$y = -2 - 1$$

$$= -3$$

$$(-1, -3)$$

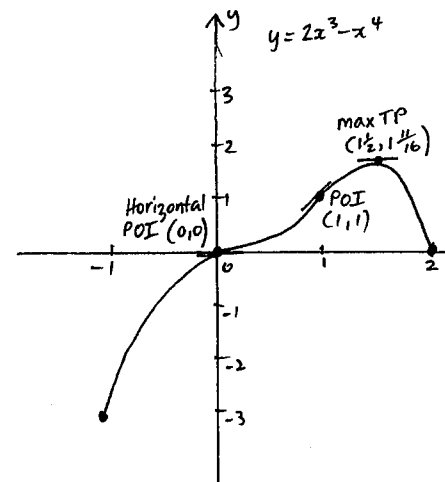
$$x = 2$$

$$y = 2 \times 8 - 2^4$$

$$= 16 - 16$$

$$= 0$$

$$(2, 0)$$



Endpoints ✓ (forgotten by many)

Stat. pts ✓

shape correct ✓

↳ Generally POIs do not have a 'kink' in them!

Comm/3

iv) the curve is concave up
 $0 < x < 1$

This was an easy question that was very poorly done!

(13)

6) b) Volume about the x axis

$$V = \pi \int_a^b y^2 dx$$

$$= \pi \int_{\pi/3}^{\pi/2} (2\sec \frac{x}{2})^2 dx$$

$$= \pi \int_{\pi/3}^{\pi/2} (4\sec^2 \frac{x}{2}) dx$$

$$= 4\pi \left[\frac{1}{2} \tan \frac{x}{2} \right]_{\pi/3}^{\pi/2}$$

$$= 4\pi \left[2 \tan \frac{x}{2} \right]_{\pi/3}^{\pi/2}$$

$$= 8\pi \left\{ \tan \frac{\pi/2}{2} - \tan \frac{\pi/3}{2} \right\}$$

$$= 8\pi \left\{ \tan \frac{\pi}{4} - \tan \frac{\pi}{6} \right\}$$

$$= 8\pi \left\{ 1 - \frac{1}{\sqrt{3}} \right\} u^3$$

Not particularly well done

* Many incorrectly squared $(2\sec \frac{x}{2})^2$ to $4\sec^2 \frac{x}{4}$ which meant they eventually could not find a neat exact form of the volume.

* Many forgot to divide by the $\frac{1}{2}$

* Note $\frac{\pi/2}{2} = \frac{\pi}{4}$,
not $\frac{\pi}{2} \times 2 = \pi$

(Calc/3)

Q7 a) $9^x - 4 \cdot 3^x + 3 = 0$
 $(3^x)^2 - 4 \cdot 3^x + 3 = 0$
 Let $m = 3^x$

$$m^2 - 4m + 3 = 0$$

$$(m-3)(m-1) = 0$$

$$m = 3$$

$$m = 1$$

$$3^x = 3$$

$$3^x = 1$$

$$x = 1$$

$$x = 0$$

This question was really well done. Good work!

Q7: Reas/2
Comm/4
Calc/2

b) P(x,y) R(-1,0) S(2,0)

i)

$$PR = 2PS$$

$$PR^2 = 4PS^2$$

$$(x+1)^2 + (y-0)^2 = 4[(x-2)^2 + (y-0)^2] \quad \checkmark$$

$$x^2 + 2x + 1 + y^2 = 4x^2 - 16x + 16 + 4y^2$$

$$3x^2 - 18x + 3y^2 + 15 = 0 \quad \checkmark$$

$$x^2 - 6x + y^2 + 5 = 0$$

Read the sentence carefully. It says that $PR = 2PS$.

Don't forget to square the 2, and don't try to fudge the answer.

(Reas/2)

ii)

$$x^2 - 6x + 9 + y^2 = -5 + 9$$

$$(x-3)^2 + y^2 = 4 \quad \checkmark$$

$\therefore$ the locus is a circle with centre (3,0) and radius 2. $\checkmark$

When completing the square, don't forget to add to both sides.

(Comm/2)

c) i) $2\sin x - 1 = 0$ for $0 \leq x \leq 2\pi$

$$\sin x = \frac{1}{2}$$

sin is positive in Q1 & Q2

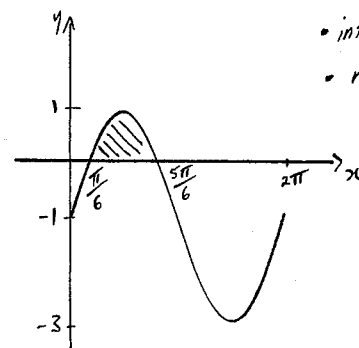
$$\therefore x = \frac{\pi}{6}, \pi - \frac{\pi}{6}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6} \quad \checkmark \text{ both. } \checkmark$$

Well done.

Use your calculator to check the angles so that you have it correct.

ii)



must show
 • intercepts & domain $\checkmark$
 • range $\checkmark$

Range is $-3 \leq y \leq 1$
 Period is 2π .

If you got this wrong, practise as many trig. graphs as you can. This is a standard question.

(comm/2)

iii) $A = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (2\sin x - 1) dx$

$$= \left[-2\cos x - x \right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$$

$$= \left\{ -2\cos \frac{5\pi}{6} - \frac{5\pi}{6} \right\} - \left\{ -2\cos \frac{\pi}{6} - \frac{\pi}{6} \right\}$$

$$= \left\{ -2 \times \frac{-\sqrt{3}}{2} - \frac{5\pi}{6} \right\} - \left\{ -2 \times \frac{\sqrt{3}}{2} - \frac{\pi}{6} \right\}$$

$$= \sqrt{3} - \frac{5\pi}{6} + \sqrt{3} + \frac{\pi}{6}$$

$$= 2\sqrt{3} - \frac{5\pi}{6} + \frac{\pi}{6}$$

$$= 2\sqrt{3} - \frac{2\pi}{3} \quad u^2 \quad OR \quad = 1.37u^2$$

Lots of students could integrate properly but then couldn't evaluate. You must be able to evaluate exact trig. ratios in any quadrant.

$$\frac{\cos \frac{\pi}{6} \text{ (quad 1)}}{= \frac{\sqrt{3}}{2}}$$

$$\frac{\cos \frac{5\pi}{6} \text{ (quad 2)}}{= -\frac{\sqrt{3}}{2}}$$

Calc 2

Q8
a) i) limiting sum when $-1 < r < 1$

since $r = x - 2$

$$-1 < x - 2 < 1$$

$$1 < x < 3$$

Q8: Reas / 3
Comm / 4
Calc / 4

- many students know $|r| < 1$ but didn't know how to solve $-1 < x - 2 < 3$

- many silly errors as well

Reas / 1

ii) $S_{\infty} = \frac{a}{1-r}$

$$2 = \frac{1}{1-(x-2)}$$

$$2 = \frac{1}{3-x}$$

$$6 - 2x = 1$$

$$-2x = -5$$

$$x = \frac{5}{2}$$

- done fairly well

Reas / 2

b) i) $f(x) = \frac{e^x}{x}$

domain: all real x where $x \neq 0$ ✓

ii) $u = e^x \quad v = x$
 $u' = e^x \quad v' = 1$

$$f'(x) = \frac{x e^x - e^x}{x^2}$$

$$= \frac{e^x(x-1)}{x^2}$$

- many students didn't write "all real x "

- done well and clearly set out

Calc / 1

iii) Stat. pt. when $f'(x) = 0$

$$\frac{e^x(x-1)}{x^2} = 0$$

$$e^x(x-1) = 0$$

$$e^x = 0 \text{ has no solution}$$

$$x - 1 = 0$$

$$x = 1$$

at $x = 1, y = e$

test nature at $(1, e)$

x	$\frac{1}{2}$	1	2
$f'(x)$	-3.3	0	+1.8

$\therefore$ a minimum turning point at $(1, e)$

- quite a few students forgot to comment about $e^x = 0$ has no solution

- testing of the stationary point was done fairly well

- just be careful to pick points near the stat. pt. a few students picked $x = 0$ and $x = -1$ to test which is too far from $x = 1$

if they pick < 0 don't give mark

Calc / 3

iv) $e^x \neq 0$ for all real x and $x \neq 0$

hence $\frac{e^x}{x} \neq 0$ for all real x

most students needed to say $e^x > 0$

- there was too much focus on the denominator

Comm / 1

v) as $x \rightarrow -\infty$
 $e^x \rightarrow 0$
 $\therefore \frac{e^x}{x} \rightarrow 0$ ✓

- done well

Comm/1

vi)

$f(x) = \frac{e^x}{x}$

Must show

- asymptote at $x=0$
- correct shape
- stat. pt. at $(1, e)$ min.

- very poorly done.

It was surprising that students who did the algebra well couldn't convert it into graphing the function correctly

Comm/2

Q9

a) $y = x^2 - 6x + 10$
 $x^2 - 6x = y - 10$
 $x^2 - 6x + 9 = y - 1$
 $(x-3)^2 = y - 1$
vertex $(3, 1)$ ✓
f.l. = $\frac{1}{4}$ $\therefore$ focus $(3, 1\frac{1}{4})$ ✓

Q9: Reas/4
Calc/4

Not well done, you must be able to convert this to parabola form
 $(x-h)^2 = 4a(y-k)$
then from this get the focal length, vertex, focus.

bi) $\int_1^3 5^{2x} dx$ $h = \frac{3-1}{4} = \frac{1}{2}$

x	f(x)	factor	f(x) × factor
1	25	1	25
1.5	125	4	500
2	625	2	1250
2.5	3125	4	12500
3	15625	1	15625

total 29900

$\int_1^3 5^{2x} dx \doteq \frac{1}{\frac{1}{2}} \times 29900$
 $\doteq 4983.333$ ✓

✓ or correct substitution into formula

Well Done

- be careful with your calculations and substitution into the formula

Calc/2

ii) $V = \pi \int_1^3 y^2 dx$ $= \pi \int_1^3 5^{2x} dx$ ✓ $= \pi \times 4983.333$ $= 15655.603 \text{ v}^3$ ✓	Not bad! Use your solution from part (i) and multiply by π <u>Calc/2</u>
c) win lose roll again 0, 1 4, 5 2, 3 $P = \frac{4}{9}$ $P = \frac{1}{6}$ $P = \frac{7}{18}$ i) $P(\text{win on 1st roll}) = \frac{4}{9}$ ✓	Not good - to win you must roll a difference of 0 or 1 there are 16 ways of doing this out of 36 <u>Reas/1</u>
ii) $P(\text{second throw}) = P(\text{roll again})$ $= \frac{7}{18}$ ✓	- A second throw needed means you must roll again ie a 2 or 3 on the first roll = $\frac{6}{36} = \frac{1}{6}$ <u>Reas/1</u>
iii) $P(\text{win 1st, 2nd, 3rd}) = P(w) + P(r, w) + P(r, r, w)$ ✓ $= \frac{4}{9} + \left(\frac{7}{18} \times \frac{4}{9}\right) + \left(\frac{7}{18} \times \frac{7}{18} \times \frac{4}{9}\right)$ ✓ $= \frac{4}{9} + \frac{4}{9} \times \frac{7}{18} + \frac{4}{9} \times \left(\frac{7}{18}\right)^2$	- write out what each roll must be and write it out - leave unsimplified <u>Reas/2</u>
iv) $P(\text{winning}) = \frac{4}{9} + \frac{4}{9} \times \frac{7}{18} + \frac{4}{9} \times \left(\frac{7}{18}\right)^2 + \frac{4}{9} \times \left(\frac{7}{18}\right)^3 + \dots$ Probability forms a limiting sum with $a = \frac{4}{9}$, $r = \frac{7}{18}$ ✓ $\therefore S_{\infty} = \frac{\frac{4}{9}}{1 - \frac{7}{18}}$ $= \frac{8}{11}$ ✓	- following on from part (iii) you should get a series with a limiting sum. <u>Reas/2</u>

Q10

(19)

ai) $\angle AED = \theta + \alpha$ (exterior $\angle$ of $\triangle BCD =$ sum of 2 opposite interior $\angle$ s) ✓

$\angle AED = \theta + \alpha$ ($\angle$ s opposite sides in an isosceles $\triangle$ are equal) ✓

$\angle AED = \angle EAB + \angle ABE$ (ext. $\angle$ of $\triangle ABE =$ sum opp. int $\angle$ s) ✓

$$\begin{aligned}\therefore \angle EAB &= \angle AED - \angle ABE \\ &= \theta + \alpha - \theta \\ &= \alpha\end{aligned}$$

must have
• correct reasons
• show $\angle EAB = \alpha$

There is more than one way to do this question. You must make sure your reasons are clear and concise. You must have a clear reason for each of the 3 steps. Learn the correct terminology.

(Comm/3)

Here's another way you could do it.

$$\begin{aligned}\angle BDC &= 180^\circ - (\angle DBC + \angle DCB) \\ &= 180^\circ - (\theta + \alpha) \quad (\text{Angle sum } \triangle BDC = 180^\circ)\end{aligned}$$

$$\begin{aligned}\angle BDA &= 180^\circ - \angle BDC \\ &= 180^\circ - (180^\circ - (\theta + \alpha)) \quad (\text{Angle sum of a straight line is } 180^\circ) \\ &= 180^\circ - 180^\circ + \theta + \alpha \\ &= \theta + \alpha\end{aligned}$$

$$\begin{aligned}\angle AED &= \angle EDA \quad (\text{Angles opposite equal sides in an isosceles } \triangle \text{ are equal.}) \\ &= \theta + \alpha\end{aligned}$$

$$\begin{aligned}\angle BEA &= 180^\circ - \angle AED \\ &= 180^\circ - (\theta + \alpha) \quad (\text{Angle sum of a straight line is } 180^\circ) \\ &= 180^\circ - \theta - \alpha\end{aligned}$$

$$\begin{aligned}\angle EAB &= 180^\circ - (\angle BEA + \angle ABE) \quad (\text{Angle sum } \triangle ABE = 180^\circ) \\ &= 180^\circ - (180^\circ - \theta - \alpha + \theta) \\ &= 180^\circ - 180^\circ + \theta + \alpha - \theta \\ &= \alpha\end{aligned}$$

Q10: Reas/18
Comm/3
Calc/1

(20)

Q10 a continued

ii) $\angle ABE = \angle CBD$ (given)
 $\angle BAE = \angle BCD$ (proven in (i))
 $\therefore \triangle ABE \equiv \triangle CBD$ (equiangular) ✓

This part can be done independent of part (i). Please use the word equiangular not equilateral. The abbreviation (AA)

(Reas/1)

iii) $\frac{AE}{CD} = \frac{BE}{BD}$ (corr. sides in sim $\triangle$ s are in the same ratio) ✓

$$AE \times BD = BE \times CD$$

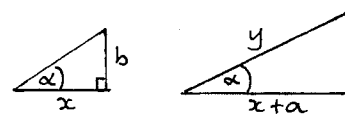
since $BD = AE$ (given) ✓

$$AE^2 = BE \times CD$$

You have to write the reason as well as the ratio to get this mark.

(Reas/2)

(b)



$$\tan \alpha = \frac{b}{x} \quad \cos \alpha = \frac{x+a}{y}$$

$$x = \frac{b}{\tan \alpha} \quad (1) \quad \therefore y = \frac{x+a}{\cos \alpha} \quad (2)$$

Substitute (1) into (2)

$$y = \frac{\frac{b}{\tan \alpha} + a}{\cos \alpha}$$

$$= \frac{b}{\tan \alpha \cos \alpha} + \frac{a}{\cos \alpha}$$

$$= \frac{b}{\frac{\sin \alpha}{\cos \alpha} \times \cos \alpha} + a \sec \alpha$$

$$= \frac{b}{\sin \alpha} + a \sec \alpha$$

$$= b \operatorname{cosec} \alpha + a \sec \alpha$$

(Reas/2)

ii) $y = a \sec \alpha + b \operatorname{cosec} \alpha$ (21)

$$= \frac{a}{\cos \alpha} + \frac{b}{\sin \alpha}$$

$$= a(\cos \alpha)^{-1} + b(\sin \alpha)^{-1}$$

using the chain rule

$$\frac{dy}{d\alpha} = -a(\cos \alpha)^{-2} \times -\sin \alpha - b(\sin \alpha)^{-2} \times \cos \alpha$$

$$= \frac{a \sin \alpha}{\cos^2 \alpha} - \frac{b \cos \alpha}{\sin^2 \alpha}$$

Common denominator.

$$= \frac{a \sin^3 \alpha - b \cos^3 \alpha}{\sin^2 \alpha \cos^2 \alpha}$$

The chain Rule is.

$$\frac{d}{dx} (f(x))^n = n f(x)^{n-1} \times f'(x)$$

Note that

$\begin{cases} a \text{ is a constant} \\ \alpha \text{ is a variable.} \end{cases}$

If you use the quotient rule on each part

$$\text{eg } \frac{d}{d\alpha} \left(\frac{a}{\cos \alpha} \right) \quad u=a \quad v=\cos \alpha$$

$u'=0 \quad v'=-\sin \alpha$
Because a is a constant its derivative is 0 not 1.

(Calc/1)

This part was done well by those who attempted it.

(Reas 1)

iii) $\frac{a \sin^3 \alpha - b \cos^3 \alpha}{\sin^2 \alpha \cos^2 \alpha} = 0$

$$a \sin^3 \alpha = b \cos^3 \alpha$$

$$\frac{\sin^3 \alpha}{\cos^3 \alpha} = \frac{b}{a}$$

$$\tan^3 \alpha = \frac{b}{a} \quad \checkmark$$

$$\tan \alpha = \sqrt[3]{\frac{b}{a}}$$

iv) min. when $\tan \alpha = \sqrt[3]{\frac{b}{a}}$ (22)

$$y = a \sec \alpha + b \operatorname{cosec} \alpha$$

$$\text{since } \sec^2 \alpha = \tan^2 \alpha + 1$$

$$\sec \alpha = \sqrt{\tan^2 \alpha + 1}$$

$$\text{also } \operatorname{cosec}^2 \alpha = 1 + \cot^2 \alpha$$

$$\operatorname{cosec} \alpha = \sqrt{1 + \cot^2 \alpha}$$

$$\therefore y = a \sqrt{1 + \tan^2 \alpha} + b \sqrt{1 + \cot^2 \alpha}$$

$$\text{since } \tan^2 \alpha = \left(\left(\sqrt[3]{\frac{b}{a}} \right) \right)^2 \quad \cot^2 \alpha = \left(\frac{b}{a} \right)^{-\frac{2}{3}}$$

$$\therefore \tan^2 \alpha = \left(\frac{b}{a} \right)^{\frac{2}{3}} \quad \cot^2 \alpha = \left(\frac{a}{b} \right)^{\frac{2}{3}}$$

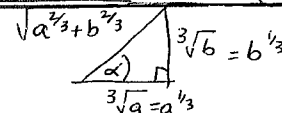
$$\therefore y = a \sqrt{1 + \left(\frac{b}{a} \right)^{\frac{2}{3}}} + b \sqrt{1 + \left(\frac{a}{b} \right)^{\frac{2}{3}}} \quad \checkmark$$

Congratulations if you made it this far and got this question out!

(Reas 2)

Here's another way to do iv)

$$\tan \alpha = \frac{\sqrt[3]{b}}{\sqrt[3]{a}}$$



$$\sec \alpha = \frac{\sqrt{a^{2/3} + b^{2/3}}}{a^{1/3}}$$

$$\operatorname{cosec} \alpha = \frac{\sqrt{a^{2/3} + b^{2/3}}}{b^{1/3}}$$

$$y = a \sec \alpha + b \operatorname{cosec} \alpha$$

$$= a \times \frac{\sqrt{a^{2/3} + b^{2/3}}}{a^{1/3}} + b \times \frac{\sqrt{a^{2/3} + b^{2/3}}}{b^{1/3}}$$

$$= a^{2/3} \sqrt{a^{2/3} + b^{2/3}} + b^{2/3} \sqrt{a^{2/3} + b^{2/3}}$$

$$= a^{2/3} \sqrt{a^{2/3} \left(1 + \left(\frac{b}{a} \right)^{2/3} \right)} + b^{2/3} \sqrt{b^{2/3} \left(\left(\frac{a}{b} \right)^{2/3} + 1 \right)}$$

$$= a^{2/3} \times a^{1/3} \sqrt{1 + \left(\frac{b}{a} \right)^{2/3}} + b^{2/3} \times b^{1/3} \sqrt{1 + \left(\frac{a}{b} \right)^{2/3}}$$

$$= a \sqrt{1 + \left(\frac{b}{a} \right)^{2/3}} + b \sqrt{1 + \left(\frac{a}{b} \right)^{2/3}}$$

Note that $\sqrt{x^2 + y^2} \neq x + y$

and $(x+y)^2 \neq x^2 + y^2$

Some basic algebraic mistakes were seen in the answers to this part.